SESSION 8

FINANCIAL PLANNING & ANALYSIS – 4

SCALER SCHOOL OF TECHNOLOGY

Learning Outcomes

Session 8: Capital Budgeting Techniques

By the end of the session, learners will be able to:

- **Calculate** the net present value (NPV) of a proposed project given its cash flows and discount rate.
- **Compute** the internal rate of return (IRR) of a project using trial-and-error/interpolation or spreadsheet functions.
- **Determine** the Payback Period and Discounted Payback Period for a given set of project cash flows..
- **Apply** the appropriate capital rationing technique (e.g., profitability index) to allocate a limited capital budget among competing projects.

Capital Budgeting: Meaning

Capital Budgeting: The process of planning and evaluating large, long-term investments, such as **purchasing new machinery, expanding the product portfolio (product line), or acquiring another business,** whose cash flows extend many years into the future.

- These decisions typically involve large sums of money committed up front
- They are largely irreversible once the investment is made (sunk cost)
- They shape the firm's future cash flows, risk profile and competitiveness for years or decades
- A single wrong capital budgeting decision can destroy far more shareholder value than an operating mistake

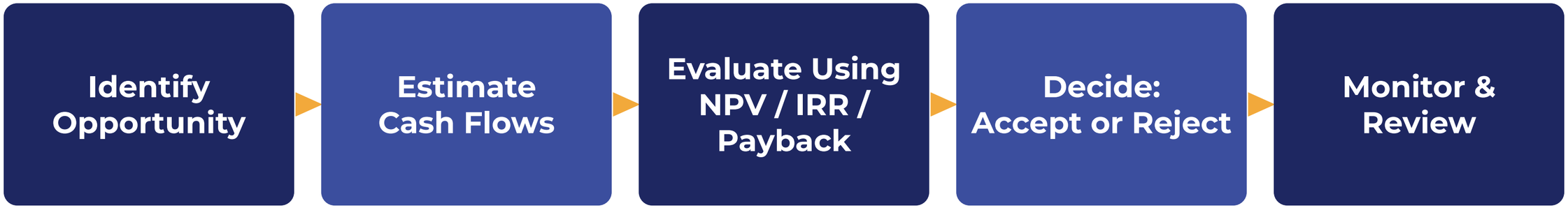
Various types of investment decisions faced by businesses:

✔ **Expansion:** Adding new capacity to grow the business. For example, Tata Motors' setting up a new EV plant at Sanand, Gujarat.

✔ **Replacement of existing assets**

✔ **Diversification:** Entering new products or markets to diversify and reduce the risk. For example, Reliance Industries entering the retail business.

Capital Budgeting: Process



Net Present Value (NPV)

NPV discounts every future cash flow back to today at the **firm's cost of capital,** then **deducts** the **initial investment.** It tells you, in today's rupees, how much value a project adds.

$$NPV = -I_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \ldots + \frac{CF_n}{(1+r)^n}$$

Where:

- I_0 = Initial investment
- $CF_1, CF_2, \ldots CF_n$ = Future cash flows
- r = Required rate of return
- n = Project life

Net Present Value (NPV)

NPV discounts every future cash flow back to today at the **firm's cost of capital,** then **deducts** the **initial investment.** It tells you, in today's rupees, how much value a project adds.

$$NPV = \Sigma \ CF_t \div (1 + r)^t - \text{Initial Investment}$$

Sum of the present value of every future cash flow, minus what you pay today

- **CF_t = expected cash flow in period t; r = cost of capital; t = time period**
- A **positive NPV** means the project is expected to **increase the value of the firm** and should be **accepted.**
- A **negative NPV** means the project destroys value and should be **rejected.**

Net Present Value (NPV)

NPV > 0	Accept — the project adds value beyond the required return
NPV = 0	Indifferent — the project earns exactly the required return
NPV < 0	Reject — the project destroys value

- NPV directly measures rupees of value created — the same metric shareholders care about
- NPVs of independent projects can simply be added together (the "value-additivity" principle)
- For ranking mutually exclusive projects or projects under a limited budget, NPV is the most reliable single criterion

Net Present Value (NPV)

Example: Bharat AutoParts Ltd.

Bharat AutoParts Ltd. is an auto-components manufacturer near Pune. It is considering installing a robotic welding line to boost quality and output.

Initial Investment	₹80,00,000 (₹80 lakh)
Expected Incremental Cash Flow	₹25,00,000 per year, for 5 years
Company's Cost of Capital (discount rate)	12% per annum

We'll use this one case throughout the session to apply every technique — NPV, IRR, Payback and Discounted Payback.

Net Present Value (NPV)

Example: Bharat AutoParts Ltd.

STEP-BY-STEP

- PVIFA (12%, 5 years) = $[1 - (1.12)^{-5}] \div 0.12 = 3.6048$
- PV of inflows = ₹25,00,000 × 3.6048
- PV of inflows = ₹90,11,938
- NPV = ₹90,11,938 − ₹80,00,000

NPV = ₹10,11,938

Positive NPV — the project is expected to add value

DECISION

Since **NPV > 0**, Bharat AutoParts should ACCEPT the robotic welding line. Investing ₹80 lakh today is expected to create roughly ₹10.12 lakh of additional value for the firm, over and above the 12% return already required by investors.

This is the same ₹80 lakh investment we will revisit under IRR, Payback and Discounted Payback.

Internal Rate of Return (IRR)

The **IRR** is the discount rate at which a **project's NPV becomes exactly zero** — in other words, the **project's own "break-even" rate of return.**

$$0 = \Sigma\ CF_t \div (1 + IRR)^t - \text{Initial Investment}$$

Solve for the rate (IRR) that sets NPV to zero

- Usually found by trial-and-error, interpolation, or a financial calculator / spreadsheet IRR() function
- **Decision rule:**

✔ **Accept the project if IRR > cost of capital;**

✔ **Reject the project if IRR < cost of capital**

$$IRR = L + \frac{NPV_L}{NPV_L - NPV_H} \times (H - L)$$

Where:

- L = Lower discount rate
- H = Higher discount rate
- NPV_L = NPV at lower rate
- NPV_H = NPV at higher rate

Internal Rate of Return (IRR)

Example: Bharat AutoParts Ltd.

Trial and error: test discount rates until NPV lands at (or near) zero.

Discount Rate	PV of Inflows (₹25,00,000/yr × 5)	NPV
12%	₹90,11,938	+ ₹10,11,938
17%	₹80,02,000	+ ₹2,000 (≈ zero)
18%	₹78,17,930	− ₹1,82,070

IRR = 17% (the rate at which NPV crosses zero)

Since IRR (17%) > cost of capital (12%), the same ACCEPT decision follows — consistent with the positive NPV.

Internal Rate of Return (IRR)

When NPV and IRR Disagree — A Hotel Chain's Dilemma

A hotel chain can **fund only ONE** of these two **mutually exclusive** energy projects:

Project A — Rooftop Solar Plant

Investment:	₹1,00,00,000
Cash Flow:	₹28,00,000 / yr for 6 yrs
Cost of Capital:	10%

Project B — LED Lighting Retrofit

Investment:	₹10,00,000
Cash Flow:	₹4,50,000 / yr for 4 yrs
Cost of Capital:	10%

Internal Rate of Return (IRR)

Comparison between NPV & IRR

	Project A (Solar)	Project B (LED Retrofit)
NPV @ 10%	₹21,94,731	₹4,26,439
IRR	=17.2%	=28.5%
Ranked #1 by…	NPV	IRR

- Project B has the higher IRR, but Project A creates far more total value (higher NPV) because it operates at a much larger scale
- Since NPV directly measures rupees of shareholder value created, NPV should win when the two rankings conflict
- Rule of thumb: for a single, independent project, use either NPV or IRR — but for mutually exclusive choices, always let NPV decide

Internal Rate of Return (IRR)

Why NPV and IRR Can Disagree

1

Scale Differences

IRR is a percentage and ignores the size of the investment; a small project can post a very high IRR while adding little total value.

2

Reinvestment Assumption

IRR implicitly assumes cash flows are reinvested at the IRR itself — often unrealistic — while NPV assumes reinvestment at the cost of capital.

3

Multiple IRRs

Projects with unconventional cash flows (e.g. a large cash outflow for decommissioning at the end) can have more than one IRR, making the rule ambiguous.

Payback Period

The **payback period** is the time it takes for a **project's cumulative cash flows to recover the initial investment** — a simple test of liquidity and risk exposure.

Payback = Years before recovery + Remaining amount ÷ Next year's cash flow

- Widely used by companies and startups because it is easy to explain to non-finance stakeholders
- **Ignores the time value of money** — every rupee is treated as equally valuable regardless of when it arrives
- **Ignores all cash flows** that occur after the payback point is reached

Payback Period

Example: Payback for Bharat AutoParts

Year	Cash Flow (₹)	Cumulative Cash Flow (₹)
1	25,00,000	25,00,000
2	25,00,000	50,00,000
3	25,00,000	75,00,000
4	25,00,000	1,00,00,000

- The ₹80,00,000 investment lies between Year 3 (₹75,00,000 recovered) and Year 4
- Amount still to recover after Year 3 = ₹80,00,000 − ₹75,00,000 = ₹5,00,000
- Payback = 3 + (₹5,00,000 ÷ ₹25,00,000) = 3.2 years

Payback Period = 3.2 years (about 3 years, 2.4 months)

Discounted Payback Period

The **discounted payback period** fixes payback's biggest flaw**: it discounts each cash flow at the cost of capital** (12%) before accumulating them.

Year	Cash Flow (₹)	Discounted CF @12% (₹)	Cumulative Discounted CF (₹)
1	25,00,000	22,32,143	22,32,143
2	25,00,000	19,93,000	42,25,143
3	25,00,000	17,79,464	60,04,607
4	25,00,000	15,88,807	75,93,414
5	25,00,000	14,18,578	90,11,992

At the end of Year 4, only ₹75,93,414 has been recovered — still short of ₹80,00,000. Remaining ₹4,06,586 ÷ Year 5's discounted CF of ₹14,18,578 = 0.29

Discounted Payback ≈ 4.29 years

*Notice: 4.29 years (discounted) is meaningfully longer than 3.2 years (simple) — **simple payback understates true recovery time.***

Investment Appraisal Techniques: Comparison

Techniques	Time Value	Key strengths	Key limitations
NPV	Yes	Directly measures ₹ value created; values are additive	Requires an accurate discount rate
IRR	Yes	Intuitive single percentage figure	Scale-blind; can conflict with NPV; multiple IRRs possible
Payback	No	Simple; highlights liquidity risk	Ignores time value and post-payback cash flows
Discounted Payback	Yes	Fixes payback's time-value flaw	Still ignores cash flows after the payback point

Capital Rationing

In practice, **firms rarely have unlimited funds.** When the **total capital budget is smaller** than the investment needed for **every positive-NPV project,** the **firm must ration capital.**

- **Soft Rationing:** a self-imposed budget ceiling set by management for planning or control reasons
- **Hard Rationing:** an external constraint — for example, lenders or capital markets are unwilling to supply more funds
- **Under rationing,** simply picking the **highest-NPV project may not be optimal.** You must **consider value created PER RUPEE invested.**

Capital Rationing: The Profitability Index

PI = PV of Future Cash Flows ÷ Initial Investment

- **The Profitability Index (PI)** measures **rupees of present value generated per rupee invested.** It **ranks projects** under a limited budget
- **PI > 1 is equivalent to NPV > 0,** Accept the project.
- When **ranking** for capital rationing, **order projects from highest to lowest PI and fund down the list until the budget runs out**

Capital Rationing: The Profitability Index

Example: Rationing a ₹1 Crore Budget

A logistics company has ₹1,00,00,000 available and is considering three independent projects (including our two earlier cases, plus a warehouse-automation proposal):

Project	Investment (₹)	NPV (₹)	Profitability Index	PI Rank
A — Solar Plant	1,00,00,000	21,94,731	1.22	2
B — LED Retrofit	10,00,000	4,26,439	1.43	1
C — Warehouse	40,00,000	5,48,960	1.14	3

- By PI ranking alone: fund B (₹10 lakh) then C (₹40 lakh) = ₹50 lakh used, total NPV = ₹9.75 lakh — leaving ₹50 lakh idle, since A needs the full ₹1 crore and is indivisible
- Funding A alone uses the entire ₹1 crore budget and delivers NPV = ₹21.95 lakh — more total value than B + C combined
- Lesson: PI ranking works well for small, divisible projects, but for large, "lumpy" (indivisible) investments, always compare total NPV across feasible combinations before deciding

Thank You!